Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow

SYLLABUS

FOR

B. Tech. Plastic Engineering

4th Year

[Effective from session 2016-17]

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW
STUDY & EVALUATION SCHEME
B. Tech. Plastic Engineering [Effective From session 2016-17]
Year 4th, SEMESTER-VII

S. No.	Course Code	Subject	Periods			Evaluation Scheme			ESE	Subject Total	Credit
			L	T	P	Sessional Exam					
						CT	TA	Total			
Theory Subjects											
1	NOE-071 to NOE-074	Open Elective I	3	1	0	30	20	50	100	150	4
2	NPL -701/ NME-701	Computer Aided Design	3	1	0	30	20	50	100	150	4
3	NPL-702	Mould & Die Manufacturing	3	1	0	30	20	50	100	150	4
4	NPL-031 to NPL-036	Departmental Elective III	3	1	0	30	20	50	100	150	4
5	NPL-041 to NPL-046	Departmental Elective IV	3	1	0	30	20	50	100	150	4
Practical / Design / Drawing											
6	NPL-751/ NME-751	CAD / CAM Lab	0	0	2	10	10	20	30	50	1
7	NPL-752	Mould & Die Manufacturing Lab	0	0	2	10	10	20	30	50	1
8	NPL-753	Industrial Training I & II Evaluation & Viva	0	0	2	-	50	50	-	50	1
9	NPL-754	Project	0	0	3	-	50	50	-	50	2
10	GP-701	General Proficiency	-	-	-	-	-	50	-	50	-
		Total	**15**	**5**	**9**					**1000**	**25**

Note- Practical Training-1 & 2 (4-weeks each) done after 4th & 6th Semesters would be evaluated in 7th semester through report and viva voce etc.

Project should be initiated in 7th semester beginning (End Semester Examination to be conducted for evaluation for 7th sem), and should be complete by the end of 8th semester with good Report and power-point Presentation etc.

Paper Code Open Electives – I
NOE-071 Entrepreneurship Development
NOE-072 Quality Management
NOE-073 Operations Research
NOE-074 Introduction to Biotechnology

Departmental Elective III
NPL-031 / NME-031: Computer Aided Manufacturing.
NPL-032 / NME-032: Project Management
NPL-033: Advanced Extrusion Process Technology.
NPL-034: Technology of Elastomers.
NPL-035: Secondary Processing Techniques.
NPL-036: Management Information System

Departmental Elective IV
NPL-041 / NME-041: Total Quality Management
NPL-042: Biodegradable Polymers.
NPL-043: Polymer Blend & Alloys.
NPL-044 / NCH 403: Mass Transfer I
NPL-045: Advance Polymeric Material.
NPL-046: Testing of Plastics Material and Product

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW
STUDY & EVALUATION SCHEME
B. Tech. Plastic Engineering [Effective From session 2016-17]
Year 4th, SEMESTER-VIII

S. No.	Course Code	Subject	Periods			Evaluation Scheme			ESE	Subject Total	Credit
			L	T	P	Sessional Exam					
						CT	TA	Total			
Theory Subjects											
1	NOE-081 to NOE-084	Open Elective II	3	1	0	30	20	50	100	150	4
2	NPL-801	Industrial Safety & Hazard Management	3	1	0	30	20	50	100	150	4
3	NPL-051 to NPL-056	Departmental Elective V	3	1	0	30	20	50	100	150	4
4	NPL-061 to NPL-066	Departmental Elective VI	3	1	0	30	20	50	100	150	4
Practical / Design / Drawing											
5	NPL-851	SEMINAR	0	0	3	-	50	50	--	50	2
6	NPL-852	PROJECT	0	0	12	-	100	100	200	300	7
7	GP-801	General Proficiency	-	-	-	-	-	50	-	50	---
		Total	**12**	**4**	**15**					**1000**	**25**

Paper Code Open Electives – II
NOE-081: Non Conventional Energy Resources
NOE-082: Nonlinear Dynamic Systems
NOE-083: Product Development
NOE-084: Automation and Robotics

Departmental Elective V
NPL-051 / NME-051: Operations Research.
NPL-052/ NMT 052: Maintenance Engineering & Management
NPL-053: Plastic Waste Management.
NPL-054: Polyurethane Technology
NPL-055: Polymer Composites.
NPL-056: Advanced Blow Moulding Process Technology.

Departmental Elective VI
NPL-061: Finite Element Method
NPL-062: Advanced Injection Moulding Technology.
NPL-063: Pneumatics & Hydraulics.
NPL-064 / NCH 501: Mass Transfer II
NPL-065: Energy Management
NPL-066: Surface Coating Technology

NPL-702: MOULD & DIE MANUFACTURING **L T P**
3 1 0

UNIT-I

Mold Making: Introduction of mold parts, Mechanism of metal cutting, types of tools, influence of tool angles, Cutting fluids, Tool materials used including coated tools. Mould material, Material selection for mould making, Properties of steels for moulds . Non-ferrous metals for moulds - Zinc base alloys and aluminium alloys, Beryllium Copper, Polyesters, Epoxies, Silicones. Review of various machining operations.

8

UNIT-II

Jig boring, Pentograph, Profile grinding, Electrical discharge machining Characteristics, physical processes, special technological features, types of EDM, design consideration & functions and technological planning. Applications of wire cut EDM in mold making. Electroforming for mold manufacturing - discussion of the process, materials for electroforming, machining for electroformed blanks

8

UNIT –III

HEAT TREATMENT PROCESSES, VARIOUS TYPES OF FURNACES.

Hobbing for mold making –

Discussion of the hobbing process & its advantages, elements of hobbing like hobbing punch, shape of the hob, materials used for cavity, lubrication, and depth of hobbing, Hobbing presses, Hobbing operations & its economy with examples.

8

UNIT –IV

Polishing technology in mold making: Definition of surface roughness, basis of polishing technology, Effect of mold materials on polishability, Types of polishing tools, Methods of polishing. Basic information on Electro sonic polishing. Principles of Electro deposition in damaged molding surfaces. Surface Texturing of molds - Process description, types of molds, types of patterns and mold shapes, metals that can be etched, mold preparation, limitations of chemical texturing.

8

UNIT-V

Computer aided Manufacturing and Measurement. Introduction to CAM; Automated Manufacturing system; Need of automation, Basic elements of automation, Levels of automation, Automation Strategies, Advantages & disadvantages of automation, classification of NC machine tools, NC Part Programming.

Manual (word address format) programming- SIMPLE Examples:Canned cycles, Subroutine, and Macro. APT programming. Geometry, Motion and Additional statements, Macro- statement Open and closed loops. Control of point to point systems-Incremental open loop control, Incremental close loop, Absolute close loop; Control loop in contouring systems; Adaptive control.

8

BOOKS-

1. "Injection Mold Design" R.G.W.Pye, East West Press Pvt. Ltd., New Delhi.
2. "Mold making handbook for Plastic Engineers", Klus Stokhert (Edt.),Hanser Publishers, NY,1983
3. "Computer Aided Engineering Design" Anupam Saxena & B. Sahay Anamaya Publishers
4. "CAD/CAM" HP Groover & EW Zimmers, Jr. Prentice Hall India Ltd.
5. "CAD/CAM" Theory and Practice Ibrahim Zeid & R Sivasubramaniam McGraw Hill

REFERENCES BOOKS-

1. "HMT Production Technology", TMH (India), 1992
2. "A New Technology", Bhattacharya, IB Publishers, 1984
3. Stoeckhert "Menning, Mold making handbook", 2nd edition, Hanser Publishers, Munich.
4. "Workshop Technology", Vol I & II, W.A.J Chapman,ELBS.
5. "Mold Engineering", Herbert Rees, Hanser Publishers, NY. George Menges & Paul Mohren
6. "How To Make Injection Molds", Hanser Publishers.
7. "Plastics Mold Engineering", DuBois; J. Harry and Pribble; W. I. (Eds.),SPE Polymer Technology

NPL-752: MOULD & DIE MANUFACTURING LAB **L T P**
0 0 2

Minimum 8 experiments

1. Study of different types of Cutting tools.
2. Letter writing on Pantograph milling
3. Study of EDM,
4. Study of Wire cut EDM
5. Study and Detailing of mould assembly
6. Manufacturing of Guide Pillar
7. Manufacturing of Pocket By Milling
8. Gas assisted and Water assisted Injection mould and Hot runner mould
9. Hand compression mould design – positive, semi positive, displacement type mould, and design with split cavities
10. Transfer mould design (pot type & top plunger type)
11. Automatic unscrewing mould
12. Study of Mould for Rotational Moulding
13. Study of Mould for Thermoforming Moulding
14. Study of Mould design for industrial component

NPL-801: INDUSTRIAL SAFETY AND HAZARD MANAGEMENT **L T P**
3 1 0

UNIT I
Industrial safety, Industrial hygiene and safety aspects related to toxicity, noise, pressure, temperature, vibrations, radiation etc. Explosions including dust, vapor, cloud and mist explosion.

8

UNIT II
Elements of safety, safety aspects related to site, plant layout, process development and design stages, identification of hazards and its estimation, risk, risk analysis and assessment methods; fault free method, event free method, scope of risk assessment, controlling toxic chemicals and flammable materials.

8

UNIT III
Toxic substances and degree of toxicity, its estimation, their entry routes into human system, their doses and responses, control techniques for toxic substances exposure, use of respirators, ventilation systems.

8

UNIT IV
Prevention of losses, pressure relief, provision for fire fighting, release of hazardous materials from tanks, pipes through holes and cracks, relief systems: types and location of relief's.

8

UNIT V
Handling, transportation and storage of flammable liquids, gases, and toxic materials and wastes, regulation and legislation, government role, risk management routines, emergency preparedness, disaster planning and management.

8

BOOKS RECOMMENDED

1. "Chemical Process Safety (Fundamentals with Applications)", D. A. Crowl and J.F. Louvar Prentice Hall (1990)
2. "Safety and Accident prevention in Chemical Operations, 2nd Edition", H.H. Fawcett and W.S. Wood John Wiley &Sons,New York, 1982.
3. "Chemical Engineering – Vol. 6" Coulson & Richardson's R.K. Sinnott, Butterworth – Heinmann Ltd., 1996.
4. "Industrial Hazards & Plant Safety", Sanjoy Banerjee,Taylor & Francis Group

DEPARTMENTAL ELECTIVE III

NPL 031 / NME-031: Computer Aided Manufacturing.
NPL 032 / NME-032: Project Management
NPL 033: Advanced Extrusion Process Technology.
NPL 034: Technology of Elastomers.
NPL 035: Secondary Processing Techniques.
NPL-036: Management Information System

NPL 033: ADVANCE EXTRUSION PROCESS TECHNOLOGY **L T P**
3 1 0

UNIT I
EXTRUSION LINES (with calibration system and takeoff / haul off units) Dies**,** Pipes**,** Corrugated pipes, Nylon braided PVC hose pipe, PVC reinforced PVC suction hose pipe, Post extrusion process
BLOWN FILM EXTRUSION
Dies, Tubular film extrusion, Blown film Co – extrusion, Polymers, combinations and properties, Advantages.
FLAT FILM & SHEETS – Dies, Introduction, Calendaring mechanisms, Polishing stack operation.

8

UNIT II
MANUFACTURE OF ORIENTED FILMS
Advantages, Process, Temperature control of film System, Stretching units
EXTRUSION COATING & LAMINATION
Dies, Introduction & Advantages, Paper film & foil lamination, Material combination, Adhesion, Pretreatment, Co - extrusion as adhesion promoter

8

UNIT III
EXTRUSION OF FOAMED INTERMEDIATE PRODUCTS.
Dies, Manufacturing process, Raw material, Post extrusion, Single stage & multi stage process, Dies & calibrating devices.
ROD EXTRUSION OF PTFE AND UHMWPE, Description of the process, Processing machine, Extrusion dies.

8

UNIT IV
PACKAGING
Introduction & Advantage**,** Monolayer & Multilayer Packaging**,** Additives & Plastic materials**,** Equipments**,** Bag making machinery**,** Thermoform / fill / seal packaging**,** Vertical form / fill / seal packaging**,** Labeling, Printing & welding
Maintenance (Electrical & Mechanical)

PROCESS CONTROL SYSTEM – Introduction, Temperature controlling system & elements, Hydraulic Controlling system & elements, Electrical controlling system & elements, Micro process control, Type of sensors, Process control, Product control.

8

UNIT V

TECHNICAL COST ANALYSIS,

Introduction to Technical cost modeling, Cost analysis method, Direct & indirect cost elements, Fixed and variable cost elements, Extrusion molding cost model – (case study for product to be manufacture), Break even point – cycle time, Annual production volume –Comparative cost analysis – (case study), Summary of cost comparisons.

8

Books-

1. Plastic Extrusion Technology – Fisher
2. Plastics Engineering Hand Book (Spi) – Berrins
3. Polymer Extrusion- Ranwendaal
4. Plastic Extrusion Technology Hand Book- Levy
5. Hand Book Of Polymeric Foams & Foam Technology-Klempner

NPL 034: TECHNOLOGY OF ELASTOMERS **L T P**
3 1 0

UNIT-I

Sources and history of natural and synthetic elastomers, significance of structure ofelastomers. Mastication, compounding ingredients and methods of compounding Reinforcing fillers and mechanism of reinforcement of elastomers.

8

UNIT-II

Production of different grades of natural rubber from latex, modified and natural rubber derivatives, Reactions of rubber, application of latex, technically specified rubber, chemistry and technology of vulcanization.

8

UNIT-III

Manufacturing processes, properties and application of synthetic elastomers viz. styrene-butadiene rubers, Acrylonitrile-butadiene rubber, butyl rubber, polychloroprene rubber.

8

UNIT-IV

Manufacturing processes, properties and applications of ethylene-propylene rubber, polyurethane elastomers, chlorosulphonated polyethylene, polysulphide and silicon, thermoplastic elastomers.

8

UNIT-V

Industrial fabrication of rubber article such as transmission belts, hoses, tyres, purged goods, compounding and processing techniques, direct manufacture of articles from latex.

8

BOOKS –

1. Rubber toughened Plastics, American Society- C. Keith Riew.
2. Rubber Technology" Hanser Gardner Publications- John Dick

NPL 035: SECONDARY PROCESSING TECHNIQUES **L T P**
3 1 0

UNIT –I

FRP Laminates: Introduction, FRP processing methods contact moulding hand lay up, spray up method vacuum bag & pressure bag moulding, filament welding, centrifugal casting, pultrusion, matched die moulding laminates, definition of terms high, pressure laminating process, types of machinery, impregnation systems – decorative and industrial laminates, continuous high pressure laminating process, application.

8

UNIT-II

Cellular Plastics: Introduction process to create foam in resins mechanical foaming, chemical foaming, physical foaming proceses to shape and solidify foams low pressure foam moulding, high pressure foam moulding, RIM extrusion foaming, casting foams, steam chest moulding structural foam moulding applications.

8

UNIT III

Machinery & joining of Plastics: Introduction – Importance of machining methods viz. cutting, drilling, blending, filling, etc. Joining principles cohesion Principle, adhesion principle- solvent cementing, DOP cementing, welding, vibration welding, hot plate welding, ultrasonic welding, Adhesive bonding examples: Mechanical fasteners.

8

UNIT –IV

Casting Processes: Dip casting, slush casting, continuous casting, cell casting, processes and applications. Calendering – Types of Calendering systems.

8

UNIT –V

Coating Processes: Roller coating, powder coating, fluidized bed coating, electrostatic spray coating, processes and applications. Other Secondary Processes: Printing, painting, hot stamping, in mould decoration, Electro plating and vacuum metallising, decorating.

8

BOOKS-

1. Plastics Finishing and Decoration, Van Nostrand Reinhold company, New York, 1986- Donatar Satar
2. Decorating Plastics, Hanser Publishers, New York, 1986 James M. Margolis
3. Plastics engineering hand book, By society of the Plastics industry- Berins
4. Manufacturing of polymer Composites, Chapman and Hall, London (1995)- B.T.Astrom
5. Plastics Processing Data Book, Van Nostrand Reinhold, New York, 1990- Donal V.Rosato & Dominick V.Rosato
6. Plastics: Materials and Processing Practice- Hall, New Jersey, 1996- A. Brent Strong
7. Joining Plastics in Production, The Welding Institute, Cambridge, 1988- M.N.Watson.

NPL 036: MANAGEMENT INFORMATION SYSTEM **L T P**
3 1 0

UNIT-I
Organisation & Types, Decision Making, Data & information Characteristics & Classification of information, Cost & value of information, Various channels of information & MIS.

6

UNIT-II
Foundation of Information System: Introduction to Information System in Business, Fundamentals of Information System, Solving Business Problems with Information System, Concept of Balanced MIS, Effectiveness & Efficiency Criteria. Tool and Techniques of MIS- dataflow diagram, flow chart etc.

10

UNIT-III
Business application of information technology, electronic commerce, Internet, Intranet, Extranet & Enterprise Solutions, Information System for Business Operations, Information system for managerial Decision Support, Information System for Strategic Advantage.

8

UNIT-IV
Managing Information Technology, Enterprise & Global Management, Security & Ethical Challenges, Planning & Implementing Change. Reports: Various types of MIS reports, GUI & Other Presentation tools.

6

UNIT-V
Advanced concepts in information system: Enterprise Resource Planning: introduction, various modules like Human Resources, Finance, Accounting, production & logistics, supply chain management, CRM, Procurement Management System Object Oriented modelling case studies.

10

BOOKS-

1. Introduction to Information System, Mc-Graw Hill- O.Brian
2. Management Information System, TMH- O.Brian.
3. Information Systems: A Management Perspective, Addison Wesley- Alter
4. Information Systems for Managers, Excel- Arora & Bhatia
5. Information System Analysis & Design, TMH- Bansal
6. Management Information System, TMH- Jawadegar
7. Information System for Modern Management, PHI- Murdick
8. Enterprise Resource Planning, TMH- Alexis Leon,

DEPARTMENTAL ELECTIVE IV

NPL 041 / NME-041: Total Quality Management (TQM)
NPL 042: Biodegradable Polymers.
NPL 043: Polymer Blend & Alloy.
NPL-044 / NCH 403: Mass Transfer I
NPL 045: Advance Polymeric Material.
NPL 046: Testing of Plastics Material And Product

NPL 042: BIODEGREDABLE POLYMERS **L T P**
3 1 0

UNIT I
CHEMISTRY AND BIOCHEMISTRY OF POLYMER DEGRADATION
Introduction, enzymes – enzyme nomenclature – enzyme specificity – physical factors affecting the activity of enzymes – enzyme mechanism, Chemical degradation initiates biodegradation, Hydrolysis of synthetic biodegradable polymers.

8

UNIT II
PARTICULATE STARCH BASED PRODUCTS
Development of Technology, Current objectives, relative starch technology, Manufacture of master batch, Conversion technology – processing precautions – moisture and temperature – rheological considerations, cyclic conversion process, physical properties of products – sample preparation – physical testing methods – test results, Quality control testing of degradation – auto oxidation measurement – biodegradation assessment – soil burial test.

8

UNIT III
BIOPOLYESTERS
Introduction, History, biosynthesis, Isolation – solvent extraction - sodium hypo chloride digestion, enzymatic digestion, Properties – crystal structure – nascent morphology, degradation - Intracellular biodegradation - extra cellular biodegradation – thermal degradation – hydrolytic degradation – environmental degradation – effects of recycling, applications, economics, future prospects.

8

UNIT IV
RECYCLING TECHNOLOGY FOR BIODEGRADABLE PLASTICS
Introduction, conventional recycling – economic incentive – recycling problems, degradable complicate recycling – polyethylene/starch film, reprocessing polyethylene/corn starch film scrap – learning to reprocess PE/S - Calcium oxide moisture scavenger – temperature control – accounting for pro-oxidant – handling PE/S repro – economics of in-plant recycling, Using PE/S repro – comparative study of PE/S repro on film properties, recycling other degradables.

8

UNIT V

TEST METHODS & STANDARDS FOR BIODEGRADABLE PLASTICS

Introduction, defining biodegradability, criteria used in the evaluation of biodegradable polymers, tiered systems for evaluating biodegradability, choice of environment, choosing the most appropriate methodology, description of current test methods – screening test for ready biodegradability, tests for inherent biodegradability, tests for simulation studies, other methods for assessing biodegradability – petri dish screen – environmental chamber method – soil burial tests, Test method developments for the future.

8

BOOKS-

1. Chemistry & Technology of Biodegradable Polymers Academic & Professional London 1994- G.J.L Griffin Blackie(ed.),
2. Biodegradable Plastics & Polymers, Elsevier 1994- Yoshiharu Doi, Kazuhiko Fukuda (ed.)
3. Handbook of Biodegradable polymers- Abraham J.Donb & Others (ed.)
4. Harvard Academic Publishers Australia 1997.

NPL 043: POLYMER BLENDS & ALLOYS **L T P**
3 1 0

UNIT-I

Introduction to polymer blends & alloys- Definitions and nomenclature reasons for making polymer blend – how to select blend components preparation of alloys & blends– economy of blending.

8

UNIT-II

Compatibilization and reactive blending: Introduction – compatibilization mechanisms – compatibilization methods compatibilization by addition of copolymer reactive blending – future trends. Determination of compatibility behaviour of Polymer solution blends by Ultrasonic interferometry.

8

UNIT- III

Rheology of polymer blends, Introduction to miscibility and immiscibility, types of blends, Flow behaviour of immiscible and miscible polymer blends.

8

UNIT – IV

Techniques for studying the morphological properties of blends and alloys through optical microscopy, SEM, Atomic Force Microscopy and TEM. Specimen preparation for SEM and TEM analysis.

8

UNIT –V

Thermal analysis of polymer blends using DSC, TGA and DMA for their various thermal properties such as thermal stability and degradation behaviour, determination of melt temperature, glass transition temperature and crystallisation temperature.

8

BOOKS-

1. Commercial Polymer Blends, Chapman & Hall, London, 1998- LA. Utracki.
2. Polymer Blends and Alloys an Overview, Asian Books Pvt.Ltd, New Delhi, 2002- RP.Singh, C.K.Das, S.K. Mustafi
3. Polymer Blends, Vol.1 & 2, Academic Press, New York, 1978- D.R.Paul & Seymour Newman
4. Polymer Mixing A self- study Guide, Hanser Publishers, Munich, 1998- Chris Rauwendaal.

NPL 045: ADVANCED POLYMERIC MATERIALS **L T P**
3 1 0

UNIT-I

High tech-areas for applications of plastics. High temperature polymers.

8

UNIT-II

Polymer blends and alloys, interpenetrating polymers networks.
Introduction – basic chemistry – General properties and applications of thermoplastic elastomers.

8

UNIT-III

Polymer concretes and polymer reagents. Introduction – basic chemistry – General properties and applications of Ultra-high modulus fibres.

8

UNIT-IV

Introduction – basic chemistry – General properties and applications of Liquid crystalline polymers.
Introduction – basic chemistry – General properties and applications of
Polymeric foams.

8

UNIT-V

Introduction – basic chemistry – General properties of Polymers for low temperature and Bio-medical applications.

8

BOOKS-

1. Temperature Engg. Thermoplastics, Elsevier pub- Collyer, A.A,High
2. Silicon Based Polymers, Noyes.- Jeigler, J.M.
3. Plastics Material- Brydson, J.A
4. Commercial Polymer Blends, Chapman & Hall, London, 1998 - LA. Utracki
5. Polymer Blends and Alloys an Overview, Asian Books Pvt.Ltd, New Delhi, 2002- C.K.Das, S.K. Mustafi, RP.Singh

NPL-046: TESTING OF PLASTICS MATERIAL AND PRODUCT **L T P**
3 1 0

UNIT – I

Consideration of the importance of testing-identification of plastics – Determination of necessary manufacturing conditions – Assessment of properties of finished products in relation to service requirements. Standard and specification – National and International standards – Test specimen preparation– Preconditioning and test atmosphere.

8

UNIT – II

Identification of common plastics materials by simple tests e.g., Visual inspection, density, effects of heat, combustion and solvents, analysis with common solvents. Mechanical Properties: Density and dimensions – Hardness-tensile strength – compressive strength – strength –flexural strength-heat strength – impact strength – dynamic stress-stain properties – creep – relaxation, friction and wear-abrasion test-fatigue-burst strength-and folding endurance.

8

UNIT –III

Thermal Properties: Specific heat and thermal conductivity thermal dependant properties - thermal endurance-glass transition temperature-thermal yield tests – Heat deflection temperature – Vicat softening temperature – Marten's heat resistance test-low temperature brittle point and flexibility test-coefficient of thermal expansion-shrinkage – Thermal stability – Thermal ageing and flammability.

Permeability Properties: Water absorption – soluble and insoluble matter – chemical resistance, environmental stress cracking resistance – ageing – gas permeability – water vapour permeability and weathering.

8

UNIT –IV

Processing and flow properties-Melt flow index, relationship of melt flow index on processing behaviour and properties.

Optical Properties – Refractive index – light transmission – haze – clarity – gloss – colour guard and microscope.

Electrical properties- Effect of polymer structure , Insulation resistance-power factor-permittivity-dielectric strength-tracking resistance-arc resistance and antistatic test.

8

UNIT –V

Product testing-pipe and fittings-film and sheets, Pipe and tube, blow bottle testing and FRP based products. Factors for designing tests for newer products. Factors affecting the quality of materials and products.

8

BOOKS -

1. "Plastics testing technology hand book"- Vishu shah
2. "Hand book Plastics test methods "-Brown RP
3. "Testing and evalution of Plastics"- Mathur ,AB & Bhardwaj IS
4. "Hand book Plastics test methods"- lves GC, Mead JA & Riley MM

DEPARTMENTAL ELECTIVE V

NPL 051 / NME 051: Operations Research.
NPL 052/ NMT 052: Maintenance Engineering & Management
NPL 053: Plastic Waste Management.
NPL 054: Polyurethane Technology
NPL 055: Polymer Composites.
NPL 056: Advanced Blow Moulding Process Technology.

NPL 053: PLASTIC WASTE MANAGEMENT **L T P**
3 1 0

UNIT –I
Introduction – Sources of plastics waste – Separation technologies, viz. Sorting – Manual, automated, Density separation, Flotation, Solvent separation, Melt filtration, Separation of resin from fiber in waste FRP.

8

UNIT – II
Plastics waste management – 4 R & I approach viz. Source reduction, Reuse, Repair, Recycling, and Incineration with examples. Plastics recycling, Classification Code of practice – Primary, secondary, territory and quaternary recycling with examples – Co-extrusion and co- injection moulding – Waste plastics as fillers.

8

UNIT – III
Mechanical recycling of commonly used plastics, such as PP,PE.PET,etc. mixed waste recycling – co-extruded films waste, commingled waste Extrusion flow moulding for production of plastics lumbers, chemical recycling/feedstoch recycling processes for recovery of oil, monomer and energy – thermolytic processes. Solvolysis – process outline for PMMA, PET,etc.

8

UNIT – IV
Recycling of plastics by surface refurbishing – Application of a coating, polishing with examples – Plastics ageing – Environmental ageing – Thermal ageing – Chemical degradation – Wear and erosion.

8

UNIT – V
Biodegradable plastics – an overiew. Environmental issues, policies and legislation in India. Plastics – Energy saving, Eco-friendly – Case studies. Life cycle analysis – a model.

8

BOOKS-

1. Recycling and recovery of plastics, Hanser Publishers, New York, 1996- R.Johanner Brandrup.
2. Plastics Waste Management, Disposal Recyding and reuse, Marcel Dekker, Inc. New York, 1993- Nabil Mustafa.

3. Plastics and the Environment, Wiley Interscience, New York (2003)- Anthony L. Andrady (Ed.)

REFERENCE-

1. Plastics Recycling, Products and Processes, Hanser Publishers, New York, 1992- R.J.Ehrig.
2. Technologies in Plastics Recyding, American Chemical Society, Washington, DC 1992.

NPL 054: POLYURETHANE TECHNOLOGY **L T P**
3 1 0

UNIT I

Introduction to polyurethane- chemistry and materials of polyurethane manufacture: basic reaction, cross linking in polyurethane, important building blocks for polyurethane (isocynates, polyols, amines and additives) - The manufacturer of polyurethanes (the process, parameters and controls).

8

UNIT II

Polyurethane processing-basic design principles of polyurethane processing equipment - steps in the polyurethane processing.
Flexible foams-(production, properties and application slabstock foam, carpet backing, flexible molded foams & semirigid molded foams.
Reinforced RIM – trends in the use of RIM and RRIM.

8

UNIT III

Rigid polyurethane foams-chemistry of raw materials, manufacturing of rigid polyurethane (manufacturing of buns, panels, foaming of applications, molded rigid foams), properties, relationship between production methods and properties- application of rigid polyurethane. Polyurethane skin integral foam-production, properties and applications.

8

UNIT IV

Solid polyurethane materials- polyurethane casting systems (cast elastomers and casting resins)-thermoplastic polyurethane elastomers: productions / processing, properties and applications, polyurethane, pains, technique and coatings, adhesives builders, elastomers fibers, manufacture / processing and applications.

8

UNIT V

Determination of composition and testing of polyurethane-chemical compositions, detection methods, identification of functional groups, determinations of properties materials and products (Characterisation, physics/mechanical, temp dependence, chemical performance, combustibility) polyurethane and environment health and safety: making and using polyurethane safety.

8

BOOKS-

1 Polyurethane Hand Book, Hanser Publication Munich- Dr. Gumter Oertal (ed.)
2 The ICI Polyurethane book -published journals by ICI, John Wiley and sons NY- George woods.

NPL 055: POLYMER COMPOSITES **L T P**
3 1 0

UNIT-1
Introduction of composite material, comparison of different materials with composites- advantages and disadvantages. Principles of composite reinforcement. Effect of fibrous reinforcement on composite strength.

8

UNIT-II
Fibrous and particulate composites, Types of reinforcement such as natural, glass, carbon/graphite, aramid fibers, high strength and high modulus fibers. Surface treatment and various forms of fibers. Reinforcement of various ananofillers such as CNT's, HNT's, MMT, nanosilica etc in polymer matric for the fabrication of nano composite.

8

UNIT-III
Thermosetting and thermoplastic materials for the composites and their selection for a particular application. Evaluation of strength and failure criteria of composites, Optimum fiber reinforcement criteria, Kelly- Davies model.

8

UNIT-IV

Processing and production techniques like hand-lay-up, spray-up, bag moldings, filament winding and pultrusion- defects.

8

UNIT-V

Prepegs manufacture and characterization. Sheet moulding and dough moulding compounds and their processing, preform and resin transfer moldings. Hybrid and sandwich type composites. Composite Testing, joining and environmental effects.

8

BOOKS-

1 Mechanical Properties of Polymers and Composites", 2nd Edn., Marcel Dekker, New York (1994)- Lawrence E.Nielsen and Robert F. Landel.
2 Manufacturing of polymer Composites, Chapman and Hall, London (1995)- B.T.Astrom.
3 Advanced Composites manufacturing, John Wiley and Sons, New York (1997)- T.G.Gutowski.
4 Mechanics of Composites Materials" Script a Book Co- Jones, R.M.

NPL 056: ADVANCED BLOW MOULDING PROCESS TECHNOLOGY **L T P**
3 1 0

UNIT I
BLOW MOLDING OF IRREGULAR PARTS
Introduction, Blow molding light weight irregular shaped parts, Moving mould sections, Integral HDPE Handle lid, Double wall internally threaded lid, Drum integral Handling HDPE Ring (L-Ring), Caps and closures, Multiple parting lines, Industrial Double wall Blow Molding, Mold preparation and other considerations.

TYPE OF AUTO FEEDING MECHANISMS
Hopper loader, Screw feeders, Hopper mounting & emptying devices, Metering devices

8

UNIT II
DIES/MOULDS
Design guideline, Introduction – Die assembly and it parts – Flow of parison, Blowing of parison, Parison thickness control, Parison swell, Calculation of Die and Mandrel dimensions, Thread closer
MOLD DESIGN BASICS
Shrinkage, Parting line, Pinch off, molded neck finishing, Venting, Clamping tonnage, Part ejection, mould cooling,

8

UNIT III
INTRODUCTION TO MOULDS
Simple blow moulds, Stretched Blow moulds, Preform moulds, Mold materials
Control system for Blow molding machines. Introduction, Temperature controlling system & elements, Hydraulic controlling system &elements, Electrical controlling system & elements, Micro process control, Types of sensors, Process control, Product Control

8

UNIT IV
MAINTENANCE
Maintenance of machines, Preventive & breakdown maintenance of different Controlling mechanism, Start and shut down process, Handling of Dies, molds and accessories, Assembly of dies & molds, Dismantling of dies & molds, Cleaning of moulds and dies, Maintenance of auxiliary equipments
Introduction to Technical cost modeling, Cost analysis method, Direct & indirect cost

8

UNIT V
TECHNICAL COST ANALYSIS
Elements, Fixed and variable cost elements, Blow molding cost model – (case study for product to be manufacture), Break even point – cycle time, Annual production volume – Comparative cost analysis – (case study), Injection molding, Thermoforming , and Summary of cost comparisons

8

BOOKS-

1 Blow Moulding of Plastics – Fisher
2 Design of Blow Moulds – R. C. Batra

DEPARTMENTAL ELECTIVE VI

NPL 061: Finite Element Method
NPL 062: Advanced Injection Moulding Technology.
NPL 063: Pneumatics & Hydraulics.
NPL 064 / NCH 501: Mass Transfer II
NPL 065 : Energy Management
NPL 066: Surface Coating Technology.

NPL 061: FINITE ELEMENT METHOD **L T P**
3 1 0

UNIT-I
INTRODUCTION
Introduction to finite difference method and finite elements method, Advantages and limitations, Mathematical formulation of FEM, Different approaches in Finite Element Method - Direct Stiffness approach, simple examples, Variational approach, Elements of variational calculus - Euler Lagrange equation, Rayliegh Ritz method, Weighted Residual methods, Point Collocation method, Galarkin method - Steps involved in FEM.

8

UNIT-II
TYPES OF ELEMENTS USED
Interpolation Polynomials - Linear elements Shape function - Analysis of simply supported beam - Element and Global matrices - Two-dimensional elements, triangular and rectangular elements - Local and Natural Co-ordinate systems.

8

UNIT-III
FINITE ELEMENT FORMULATION OF FIELD PROBLEMS
1-D and 2-D heat transfer, fluid flow (incompressible and non viscous fluid) in ducts, Simple electrical and magnetic field problems. Simple Numerical examples

8

UNIT-IV
FINITE ELEMENT FORMULATION OF SOLID MECHANICS PROBLEMS
1-D problem of shaft; Truss element analysis of pinned truss, Plane stress/strain problems, Axi-symmetric problems, thin plate problems; Vibration of shafts & beams.

8

UNIT-V
NUMERICAL METHODS IN FEM
Evaluation of shape functions - One dimensional & triangular elements, Quadrilateral elements, Isoperimetric elements - Numerical Integration, Gauss Legendre quadrature - Solution of finite element equations – Gauss Elimination Method, Cholesky decomposition.

8

BOOKS-

1 The Finite Element Method O.C. Zienkiewicz and R.L. Taylor McGraw Hill
2 An Introduction to Finite Element Method J. N. Reddy McGraw Hill
3 Finite Element Procedure in Engineering Analysis K.J. Bathe McGraw Hill
4 Finite Element Analysis C.S. Krishnamoorthy Tata McGraw Hill
5 Concepts and Application of Finite Element Analysis, R.D. Cook, D.S. Malcus and M.E. Plesha John Wiley
6 Introduction to Finite Elements in Engineering T.R Chandragupta and A.D. Belegundu Prentice Hall India
7 Finite Element and Approximation O.C. Zenkiewicy & Morgan -
8 Numerical Methods E Balagurusamy Tata McGraw Hill

NPL 062: ADVANCED INJECTION MOULDING TECHNOLOGY **L T P**
3 1 0

UNIT I

INTRODUCTION TO LIGHT WEIGHTED FORMED / HOLLOW INJECTION MOLDED ARTICLES

Gas-assisted injection moulding – Introduction, Field of application, Materials – Gas injection systems & mechanisms, Processing technology & steps, Discontinuous pressure generation Continuous pressure generation, Gas pressure control, Gas Injection units (through nozzle & into mould), Mould technology, Product properties, Design guidelines & quality, Troubleshooting

INJECTION MOULDING WITH FUSIBLE CORE TECHNOLOGY

Introduction, application and advantages

8

UNIT II

INJECTION MOULDING MACHINE FOR THERMOSETS

Feeding system, Oil heated barrel, Oil temperature controller, Screw / barrel / nozzle design

CONTROLLED LOW PRESSURE INJECTION MOULDING

Concept of low- pressure moulding, Alternative injection moulding strategies, Operating conditions, Foaming agents, Equipments, Process Control, Thermoplastic solid moulding, Structural foam moulding

8

UNIT III

HYDRAULIC APPLICATION

Hydraulic system, Multi pump, Variable pump, Accumulator, High capacity valves, Intensifier, Hydraulic safety gate

INTRODUCTION AND PROCESS IN BRIEF

Co injection (sandwich) moulding, Injection Blow Moulding, Injection Stretch Blow Moulding – Resin Transfer Moulding, Injection Moulding Decoration, Assembly Techniques(introduction only), Moulding-injection Assembly – Chemical Bonding system, Thermal welding Method, Assembly with Fasteners, Orientation and Warpage Prediction in polymer
processing, Lamellar injection moulding (LIM) process for multi phase polymer System.

8

UNIT IV

TYPE OF MOULDS

Introduction to mould design only, Hot runner moulds, Gas – assisted injection moulds, Low– pressure injection moulds, Moulds for two - colour/ two component moulding, Moulds for rotating core process, Moulds for retractable core process, Reaction injection moulds.

INTRODUCTION TO PACKAGING PRODUCTS ITS PROCESS & PRINTING TECHNIQUES

Equipments, Auto labelling, Surface treatment, Printing techniques & it's process

8

UNIT V

TECHNICAL COST ANALYSIS

Introduction to Technical cost modeling, Cost analysis method, Direct & indirect cost elements, Fixed and variable cost elements, Injection molding cost model – (case study for product to be manufacture), Break even point – cycle time, Annual production volume, Comparative cost analysis – (case study), Summary of cost comparisons

8

BOOKS-

1. Plastic Engineering Handbook – Berrins
2. Injection Moulding Theory and Practice – Rubins

NPL 063: PNUEMATICS & HYDRAULICS **L T P**

3 1 0

UNIT I

INTRODUCTION TO FLUID POWER AND DRIVES

Fluid Power Systems – Application of fluid power – Properties of hydraulics fluids – Hydraulics pumps – Characteristics – Pump Selection -, Hydraulics Actuators – Linear, Rotary – Selection – Characteristics – Cylinder Mountings, cushioning, pipe fittings.

8

UNIT II

FLUID POWER ELEMENTS

Pressure control valves, flow control valves, directional control valves – working principle and construction, special type valves, servo valves, Cartridge valves

Actuation methods, Shock absorbers – Accumulator – Symbol for fluid power elements.

8

UNIT III

HYDRAULICS CIRCUITS

Hydraulics circuits – automatic reciprocating circuit – speed control circuit - Meter in - Meter out – Sequencing Circuits – Synchronizing circuits, Accumulator circuit – Safety circuits – Hydraulic Motor braking System. Design of Hydraulic circuits

8

UNIT IV

PNEUMATIC SYSTEMS

Pneumatic fundamentals Filter, regulator, lubricator, air motors, air cylinders, pneumatic valves, Basic Pneumatic circuits – Hydro Pneumatic Systems – air- oil cylinder, air – oil reservoir, air – oil intensifier and simple circuits

8

UNIT V

FLUID LOGIC CONTROL

Principle of fluid Logic control, Basic fluidic devices – Fluid sensors Fluidic circuits – sequencing control, continuous reciprocation, Electrical controls – electrical components – simple electro hydraulic/ pneumatic circuits. PLC application in fluid power control.

8

BOOKS-

1 Antony Esposito "Fluid Power with applications"
2 Mazumdar S. R "Pneumatic Systems – Principles and Maintenance"
3 Mazumdar S. R "Oil Hydraulics Systems – Principles and Maintenance"
4 John Pipenger & Tyler Hicks "Industrial Hydraulics"
5 Chandashekhara P. K. "Fluid Power"
6 Kripal Singh "Automobile Engineering Vol. I"

NPL 065: ENERGY MANAGEMENT **L T P**

3 1 0

UNIT-1

Introduction to energy, Sources of energy, Forms of energy, Energy reserves, renewable energy sources, Unites of energy and the laws of thermodynamics,, Energy consumption and GDP, energy database , Energy demand analysis, Costs of exploration and utilization of depletable resources, energy pricing, National energy plan.

7

UNIT-2

Energy audit concepts, Energy audit based on 1st law and 2nd law of thermodynamics, Mass and Energy balances, Availability analysis, Evaluation of energy conserving opportunities, Economic analysis and life cycle costing.

7

UNIT-3

Energy conservation areas, Energy transmission and storage, Plant wide energy optimization Models, Data base for energy management, Energy conservation through controls, Computer aided energy management, Program organization and methodology.

7

UNIT-4

Electrical energy conservation in building lighting, heating, ventilating and air conditioning, Energy efficient motor, power factor improvement in power systems, Energy audit of Combustion process, Boilers, Turbines, compressors, Pumps, Heat exchangers, Condensers, Use of industrial, wastes.

9

UNIT-5

Energy environment interaction, Environmental issues, Global warning, Carbon dioxide emissions, Depletion of ozone layer, Government's regulations, Energy economy interaction.

7

BOOKS-

1 Energy Management and condevtion, by Clive Beggs, Butterwoth- Heinemann Elsevier Science.
2 Optimising Energy Efficiency in the Industry, By Rajan, Tata Mc Graw Hill
3 Guide to energy Management, By C.L Capehart, Fairmont Press.

NPL 066: SURFACE COATING TECHNOLOGY **L T P**

3 1 0

UNIT-1

Origin and development of surface coatings. Constituents of paint, varnishes and lacquers.

6

UNIT-2

Functions of coating and mechanism of film formation. Characteristics of natural and synthetic film formation.

8

UNIT-3

Pigments and pigmentation. Dispersion techniques. Fundamentals of coating formulations based on natural and synthetic polymers. Role of wetting agents, driers, solvents and plastisizers in coatings.

9

UNIT-4

Surface preparation and pretreatments. Rheological behaviors of coatings. Application methods and curing techniques.

9

UNIT-5

Specialty coatings like water based, powder and high solid, etc. industrial and architectural finishes.

8

BOOKS-

1. Organic Coatings: Science and Technology, Zeno W. Wicks, John Wiley and Sons Ltd
2. Surfactants in Polymers, Coatings, Inks and Adhesives, Edited by D.R. Karsa,
3. Coatings Materials and Surface Coating ; Arthur A. Tracton
4. Donatas Satas, Arthur A Tracton , Coatings Technology Handbook
5. Coatings Technology: Fundamentals, Testing, and Processing Techniques Arthur A Tracton,